Complex Analysis Master Course

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Chapter 1

First Week

1.1 Lecture 1 12/10/2026

Definition 1. *the rectangular form of $f : D \subseteq \mathbb{C} \to \mathbb{C}$ is given by $f(z) = u(x,y) + iv(x,y)$, where $z = x + iy$ and $u, v : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ are real-valued functions.*

Example 1. *Consider the function $f(z) = z^2$. Then $f(z) = (x+iy)^2 = x^2 - y^2 + 2ixy$. Thus, $u(x,y) = x^2 - y^2$ and $v(x,y) = 2xy$.*

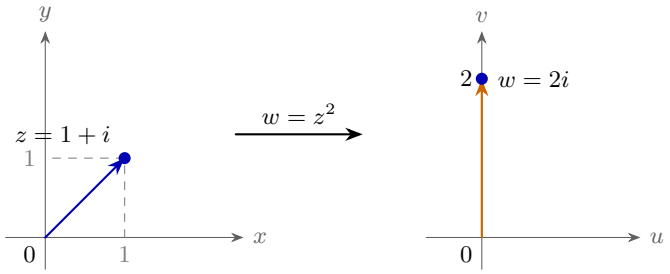


Figure 1.1: Squaring in rectangular coordinates: $(1+i)^2 = 0+2i$. The horizontal and vertical coordinates of the image are $u = x^2 - y^2$ and $v = 2xy$.

Definition 2. *the polar form of $f : D \subseteq \mathbb{C} \to \mathbb{C}$ is given by $f(re^{i\theta}) = U(r,\theta) + iV(r,\theta)$, where $z = re^{i\theta}$, $r \geq 0$, and $\theta \in [0, 2\pi)$. When $f(z) \neq 0$, one may also write $f(re^{i\theta}) = R(r,\theta)e^{i\Phi(r,\theta)}$, where $R = |f(z)|$ and Φ is an argument of $f(z)$.*

Example 2. *Consider the function $f(z) = z^2$. Then $f(z) = (re^{i\theta})^2 = r^2e^{i2\theta}$. Thus, the polar form of $f(z)$ is given by $r^2e^{i2\theta}$.*

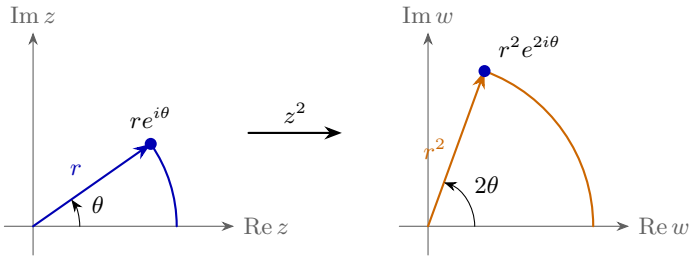


Figure 1.2: The map $z \mapsto z^2$ squares the modulus and doubles the argument (modulo 2π). Radial scales in the two panels are schematic.

Definition 3. $\lim_{z\to z_0} f(z) = w_0$ *if for every* $\epsilon > 0$, *there exists a* $\delta > 0$ *such that* $z \in D$ *and* $0 < |z - z_0| < \delta$ *imply* $|f(z) - w_0| < \epsilon$.

Remark 1. $\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$ *and* $\lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$ *if and only if* $\lim_{z\to z_0} f(z) = w_0$.

Definition 4. f *is continuous at* z_0 *if* $\lim_{z\to z_0} f(z) = f(z_0)$.

Definition 5. f *is differentiable at* z_0 *if the limit*

$$f'(z_0) = \lim_{z\to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists.

Example 3. *consder the function* $f(z) = |z|^2$ *use the definition to find the* $f'(0)$.

Solution 1. *We have* $f(z) = |z|^2 = z\bar{z}$. *Then*

$$f'(0) = \lim_{z\to 0} \frac{|z|^2}{z} = \lim_{z\to 0} \bar{z} = 0, \qquad |\bar{z}| = |z| \longrightarrow 0.$$

Thus, $f'(0) = 0$.

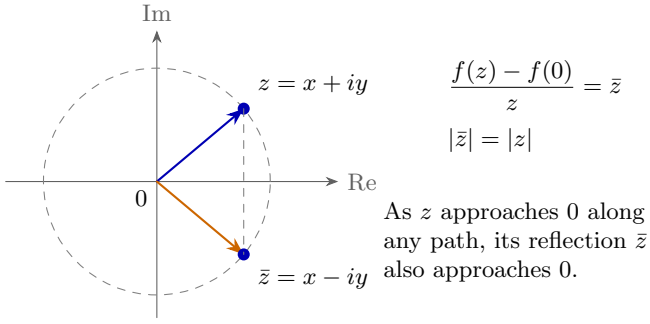


Figure 1.3: For $f(z) = |z|^2$, the difference quotient at 0 is the reflection $\bar{z}$, whose distance from 0 tends to zero. The orange vector represents the quotient, not $f(z)$.

Theorem 1. *Let $f(z) = u(x, y) + iv(x, y)$ be defined in a neighborhood of $z_0 = x_0 + iy_0$. Suppose the first partial derivatives of u and v exist and are continuous in that neighborhood. Then f is differentiable at z_0 if and only if the Cauchy–Riemann equations hold at (x_0, y_0):*

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Example 4. *Consider the function $f(z) = z^2$. Then $f(z) = (x + iy)^2 = x^2 - y^2 + 2ixy$. Thus, $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$. We have*

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = 2x, \quad \frac{\partial u}{\partial y} = -2y, \quad \frac{\partial v}{\partial x} = 2y.$$

Thus, the Cauchy-Riemann equations are satisfied and f is differentiable at every point in $\mathbb{C}$.

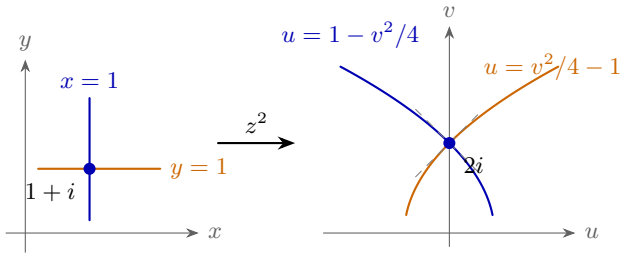


Figure 1.4: The perpendicular lines $x = 1$ and $y = 1$ map to parabolas meeting at $2i$ with perpendicular tangents. Since $f\prime(1 + i) = 2 + 2i \neq 0$, squaring preserves angles locally there; at $z = 0$ the derivative vanishes.

Example 5. *Consider the function $f(z) = \bar{z}$. Then $f(z) = x - iy$. Thus, $u(x, y) = x$ and $v(x, y) = -y$. We have*

$$\frac{\partial u}{\partial x} = 1, \quad \frac{\partial v}{\partial y} = -1, \quad \frac{\partial u}{\partial y} = 0, \quad \frac{\partial v}{\partial x} = 0.$$

Thus, the Cauchy-Riemann equations are not satisfied and f is not differentiable anywhere.

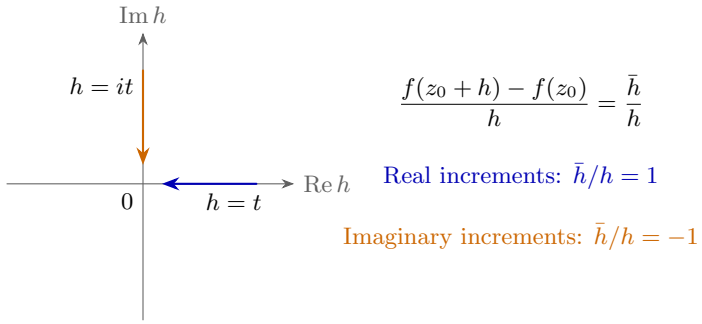


Figure 1.5: Conjugation reflects the plane across the real axis (as in Figure 1.3). At every z_0, real and imaginary increments give different difference quotients, so the complex derivative does not exist.

Definition 6. *A function f is said to be an analytic at a point z_0 if it is differentiable in a neighborhood of z_0. If f is analytic at every point in a domain D, then f is said to be analytic in D.*

Example 6. *Consider the function $f(z) = |z|^2$ differentiable at $z_0 = 0$ but not analytic at $z_0 = 0$ since it is not differentiable in any neighborhood of $z_0 = 0$.*

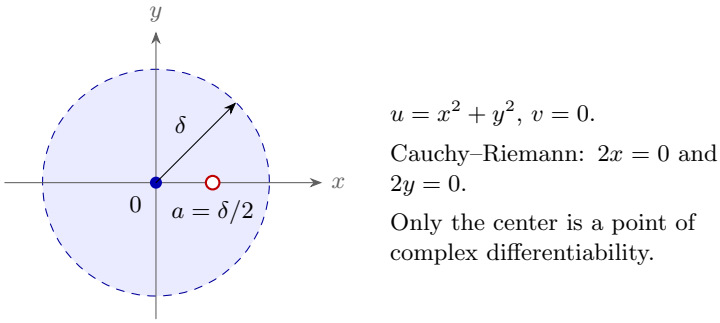


Figure 1.6: Every disk about 0 contains nonzero points such as $a = \delta/2$ where $|z|^2$ is not differentiable. The open marker here denotes such a point, not a point missing from the domain. Hence differentiability at 0 does not imply analyticity there.

Definition 7. *Consider The point z_0 is said to be a singular point of f if f is not analytic at z_0.and there exists a sequence of points $z_n \to z_0$ such that f is analytic at each z_n.*

Definition 8. *z_0 is said to be an isolated singular point of f if f is not analytic at z_0 and there exists a neighborhood of z_0 in which f is analytic at every point except z_0.*

Remark 2. *the neighborhood of z_0 in which f is analytic at every point except z_0 is called a punctured neighborhood of z_0.*

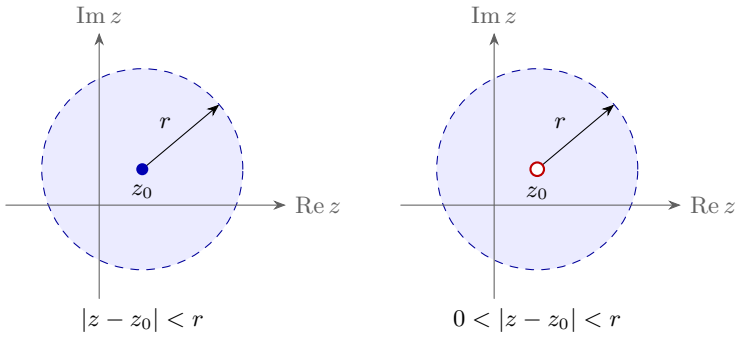


Figure 1.7: An open disk and a punctured disk in the complex plane. Dashed boundaries are excluded in both; the hollow center on the right is also excluded.

Exercise 1.

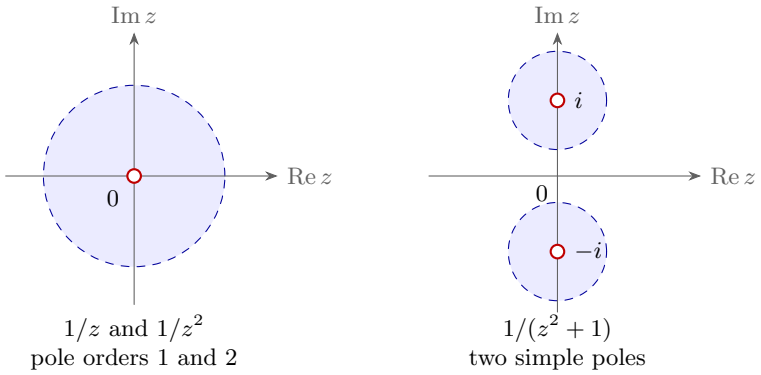
1. *Consider the function $f(z) = \frac{1}{z}$. Then f is not analytic at $z_0 = 0$ and there exists a punctured neighborhood of $z_0 = 0$ in which f is analytic at every point except $z_0 = 0$. Thus, $z_0 = 0$ is an isolated singular point of f.*

2. *Consider the function $f(z) = \frac{1}{z^2+1}$. Then f is not analytic at $z_0 = i$ and $z_0 = -i$ and there exists a punctured neighborhood of $z_0 = i$ and $z_0 = -i$ in which f is analytic at every point except $z_0 = i$ and $z_0 = -i$.*

3. *Consider the function $f(z) = \frac{1}{z^2}$. Then f is not analytic at $z_0 = 0$ and there exists a punctured neighborhood of $z_0 = 0$ in which f is analytic at every point except $z_0 = 0$. Thus, $z_0 = 0$ is an isolated singular point of f.*

4. *Consider the function $f(z) = \frac{1}{z^2+1}$. Then f is not analytic at $z_0 = i$ and $z_0 = -i$ and there exists a punctured neighborhood of $z_0 = i$ and $z_0 = -i$ in which f is analytic at every point except $z_0 = i$ and $z_0 = -i$. Thus, $z_0 = i$ and $z_0 = -i$ are isolated singular points of f.*

5. *Consider the function $f(z) = \frac{1}{\sin(\pi/z)}$. Its singularities at $z = 1/n$, $n \in \mathbb{Z} \setminus \{0\}$, accumulate at 0. Thus, $z = 0$ is a non-isolated singular point.*



$1/z$ and $1/z^2$
pole orders 1 and 2

$1/(z^2+1)$
two simple poles

Figure 1.8: Exercises 1–4: each hollow point is an isolated pole. The shaded punctured disks contain no other singularities. For the poles at $\pm i$, radius $r < 2$ suffices; the pictured disks have radius less than 1. The accumulating poles in Exercise 5 are shown in Figure 1.9.

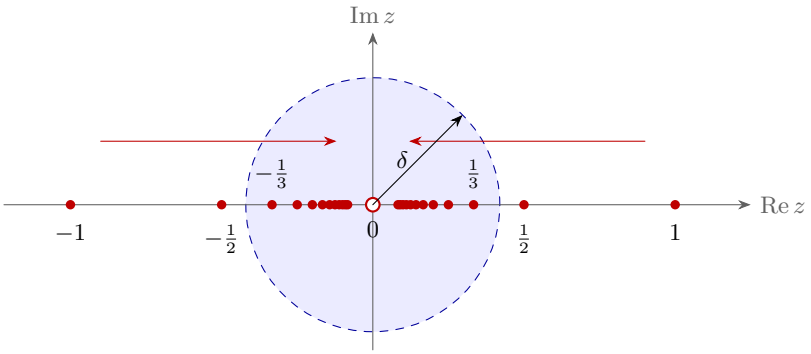
1.2 Lecture 2 14/10/2026

Example 7. $f(z) = \frac{1}{\sin(\pi/z)}$. *Determine whether $z = 0$ is a singular point (isolated or not).*

Solution 2. *The function is not defined at $z = 0$. To find its other singularities, solve*

$$\sin\left(\frac{\pi}{z}\right) = 0 \quad \Longleftrightarrow \quad \frac{\pi}{z} = n\pi, \qquad n \in \mathbb{Z}.$$

Since $z \neq 0$, the solutions are $z = 1/n$ for $n \in \mathbb{Z} \setminus \{0\}$. These are poles of f, and they accumulate at 0 because $1/n \to 0$ as $|n| \to \infty$. Consequently, every punctured neighborhood of 0 contains singularities of f. Therefore, $z = 0$ is a non-isolated singular point, not an isolated singular point.



$$|n| > 1/\delta \quad \Longrightarrow \quad 0 < |1/n| < \delta$$

Figure 1.9: The poles $z = 1/n$ approach 0 from both sides of the real axis. Only finitely many are drawn; infinitely many lie in every punctured disk about 0, so 0 is not isolated. The shaded disk illustrates one choice of δ; the inequality applies to every $\delta > 0$.

Example 8. *$f(z) = \log z$. Determine whether $z = 0$ is an isolated singularity.*

Solution 3. *The complex logarithm is multivalued: if $z = re^{i\theta}$, then*

$$\log z = \ln r + i(\theta + 2\pi k), \qquad k \in \mathbb{Z}.$$

When z travels once around 0, its argument increases by 2π, so the value of the logarithm changes by $2\pi i$. Thus, no single-valued analytic branch of $\log z$ exists on a full punctured neighborhood of 0. The point $z = 0$ is a branch point (often called a logarithmic branch point), not an isolated singularity in the sense requiring analyticity throughout a punctured neighborhood. A branch such as the principal logarithm is analytic only after a branch cut is removed.

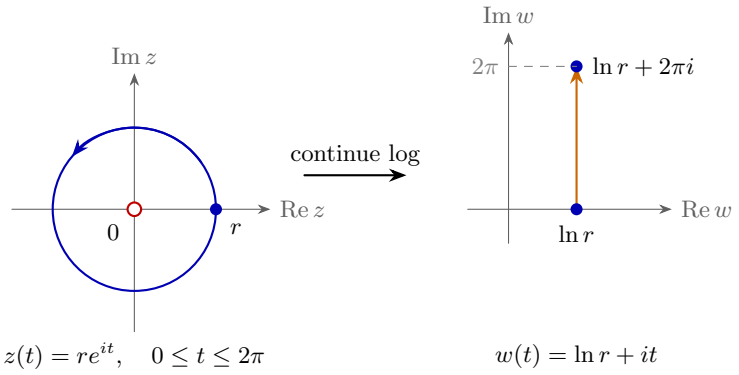


Figure 1.10: Analytic continuation of a logarithm around 0 returns to the same input r but changes the output by $2\pi i$. This is a path of continued values, not a single-valued logarithm on the full punctured plane.

Example 9. *Let $f(z) = \operatorname{Log} z$ be the principal logarithm. Determine whether $z = 0$ is an isolated singularity.*

Solution 4. *The principal logarithm is defined by*

$$\operatorname{Log} z = \ln|z| + i \operatorname{Arg} z, \qquad -\pi < \operatorname{Arg} z < \pi,$$

on the slit plane $\mathbb{C} \setminus (-\infty, 0]$. It is analytic throughout this domain. However, every disk centered at 0 meets the branch cut, so there is no full punctured neighborhood of 0 on which this branch is defined and analytic. Therefore, $z = 0$ is not an isolated singularity of the principal logarithm in the standard sense; it is a branch point. As $z \to 0$ within the slit plane, $\operatorname{Re}(\operatorname{Log} z) = \ln|z| \to -\infty$.

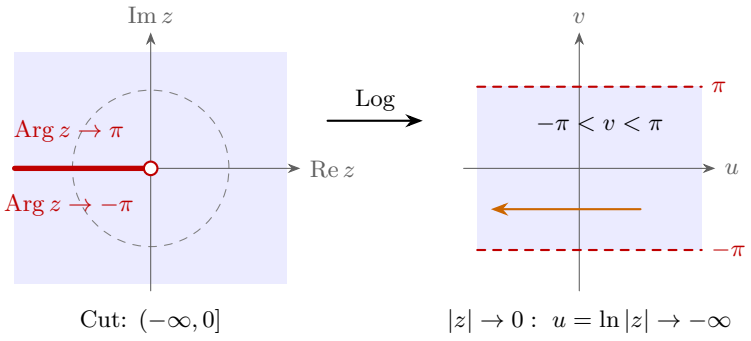


Figure 1.11: The principal logarithm maps the slit plane onto the horizontal strip $-\pi < \operatorname{Im} w < \pi$ (both extend beyond the drawing). The red cut and dashed strip edges are excluded. Every disk centered at 0 meets the cut.

1.3 Summary of Examples: Singularities

Table 1.1: Summary of the functions discussed and their singularity behavior.

Function	Point(s)	Classification	Explanation
z^2	None	No singularities; entire	It is analytic at every point of $\mathbb{C}$.
$\bar{z}$	Every point	Not analytic anywhere	It fails the Cauchy–Riemann equations everywhere, so there is no punctured neighborhood on which it is analytic.
$\lvert z\rvert^2$	0	Differentiable at 0, but not an isolated singularity	It is differentiable at 0 but not analytic in any neighborhood of 0.
$\frac{1}{z}$	0	Isolated pole (order 1)	It is analytic in a punctured neighborhood of 0.
$\frac{1}{z^2}$	0	Isolated pole (order 2)	It is analytic in a punctured neighborhood of 0.
$\frac{1}{z^2+1}$	$i, -i$	Isolated simple poles	The denominator has simple zeros at i and $-i$; each point has a punctured neighborhood containing no other singularities.
$\frac{1}{\sin(\pi/z)}$	$z = 1/n$, $n \neq 0$; accumulation at 0	Each $1/n$ is an isolated simple pole; 0 is a non-isolated singular point	The poles $1/n$ accumulate at 0, so no punctured neighborhood of 0 is free of other singularities.
$\log z$	0	Logarithmic branch point; not an isolated singularity	Analytic continuation once around 0 changes the logarithm by $2\pi i$, so no single-valued analytic branch exists on a full punctured neighborhood.
$\operatorname{Log} z$ (principal branch)	0	Branch point; not an isolated singularity	The principal branch is analytic on the slit plane $\mathbb{C} \setminus (-\infty, 0]$, not on a full punctured neighborhood of 0.

Remark 3. *An isolated singularity requires the function to be analytic on a full punctured neighborhood of the point. A branch point such as* 0 *for* $\log z$ *is therefore not an isolated singularity in this sense.*

1.4 Integration

Complex integration begins with ordinary real integrals of the real and imaginary parts. To integrate along a curve in the complex plane, we then parametrize the curve and include its complex tangent vector. Throughout this section, curves are oriented: arrows indicate the direction in which they are traversed.

1.4.1 Complex-valued functions of a real variable

Definition 9. *Let $g : [a, b] \to \mathbb{C}$ have the form $g(t) = u(t) + iv(t)$, where u and v are Riemann integrable real-valued functions. Define*

$$\boxed{\int_a^b g(t)\,dt = \int_a^b u(t)\,dt + i\int_a^b v(t)\,dt.}$$

In particular, this integral exists whenever g is continuous on $[a, b]$.

Linearity and the fundamental theorem of calculus apply componentwise. For example, if $G : [a, b] \to \mathbb{C}$ is continuously differentiable, then

$$\int_a^b G'(t)\,dt = G(b) - G(a).$$

We also have the useful inequality

$$\left|\int_a^b g(t)\,dt\right| \leq \int_a^b |g(t)|\,dt.$$

Example 10. *Evaluate* $\displaystyle\int_0^1 (t + it^2)\,dt$.

Solution 5. *Integrating the two components separately gives*

$$\int_0^1 (t + it^2)\,dt = \left[\frac{t^2}{2}\right]_0^1 + i\left[\frac{t^3}{3}\right]_0^1 = \frac{1}{2} + \frac{i}{3}.$$

The values $g(t) = t + it^2$ trace a parabolic arc in the complex plane. The integral is a complex number, represented by the vector to $\frac{1}{2} + \frac{i}{3}$; it is not the length of that arc.

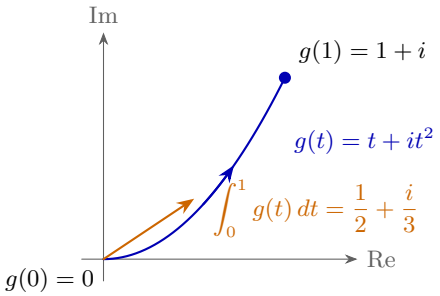


Figure 1.12: The blue arc shows the values of the integrand as t increases. The orange vector represents the value of its integral with respect to t.

1.4.2 Contours and their parametrizations

Definition 10. *Let $\gamma(t) = x(t) + iy(t)$, $a \le t \le b$, where $a < b$. The following terminology describes the path and its parametrization.*

1. *An* arc *(or curve) is a continuous map $\gamma : [a, b] \to \mathbb{C}$; equivalently, both x and y are continuous on $[a, b]$.*

2. *An arc is* simple *if it is injective: $\gamma(t_1) \neq \gamma(t_2)$ whenever $t_1 \neq t_2$. Thus it does not cross itself or retrace a point.*

3. *An arc is* closed *if $\gamma(a) = \gamma(b)$. It is* simple closed *if this endpoint identification is the only repetition: γ is injective on $[a, b)$.*

4. *A* smooth regular arc *has a continuous derivative $\gamma'(t) = x'(t) + iy'(t)$, with one-sided derivatives at the endpoints, and $\gamma'(t) \neq 0$ for $a < t < b$. The nonzero derivative supplies a tangent direction.*

5. *A* contour *is a continuous path consisting of finitely many smooth regular arcs joined end to end; it is also called a piecewise smooth path. Corners at the joins are allowed.*

The initial and terminal points are $\gamma(a)$ and $\gamma(b)$. The orientation is the direction of increasing t.

Remark 4. *Simplicity and smoothness describe different properties: simplicity concerns repeated points, while smoothness concerns derivatives. A simple closed curve is not injective on the full interval $[a, b]$, because its endpoints must coincide.*

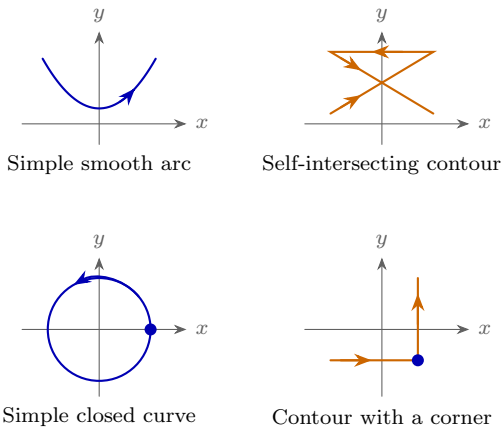


Figure 1.13: Four types of paths in the complex plane. A crossing prevents simplicity; a corner prevents a single smooth regular parametrization through that join, but piecewise smooth integration still applies. The dot on the circle marks its common initial and terminal point.

Writing $\gamma(t) = x(t) + iy(t)$, we have

$$\gamma'(t) = x'(t) + iy'(t), \qquad L(\gamma) = \int_a^b |\gamma'(t)|\, dt.$$

For a piecewise smooth contour, the integral for the length is the sum over its smooth pieces. Two standard parametrizations will be used repeatedly:

- The line segment from z_0 to z_1:

$$\gamma(t) = z_0 + t(z_1 - z_0), \qquad 0 \le t \le 1.$$

- The circle with center c and radius $R > 0$, traversed once counterclockwise:

$$\gamma(t) = c + Re^{it}, \qquad 0 \le t \le 2\pi.$$

  Restricting the angle interval gives a circular arc. Replacing e^{it} by e^{-it} gives clockwise traversal of the full circle.

1.4.3 Definition and calculation of a contour integral

Definition 11. *Let f be continuous on the image of a contour $\gamma : [a, b] \to \mathbb{C}$. The* contour integral *of f along γ is*

$$\boxed{\int_\gamma f(z)\, dz = \int_a^b f(\gamma(t))\gamma'(t)\, dt.}$$

If γ has several smooth pieces, compute the integral on each piece and add the results. For a closed contour, we may write $\oint_\gamma f(z)\, dz$.

To calculate an integral, choose a parametrization with the required orientation, substitute $z = \gamma(t)$ and $dz = \gamma'(t)\,dt$, and evaluate the resulting integral over the real parameter interval.

Remark 5. *The factor $\gamma'(t)$ is essential: $\int_a^b f(\gamma(t))\,dt$ is generally not a contour integral. Also, $dz = \gamma'(t)\,dt$ differs from the arc-length element $ds = |\gamma'(t)|\,dt$.*

For $f = u + iv$ and $z = x + iy$, multiplication gives the equivalent formula

$$\int_\gamma f(z)\,dz = \int_\gamma (u\,dx - v\,dy) + i\int_\gamma (v\,dx + u\,dy).$$

Here u and v are evaluated at the points $(x(t), y(t))$ of the contour.

Example 11. *Evaluate $\displaystyle\int_\gamma z\,dz$, where γ is the straight segment from 0 to $1+i$.*

Solution 6. *Set $\gamma(t) = (1+i)t$, $0 \le t \le 1$. Then $\gamma'(t) = 1 + i$, so*

$$\int_\gamma z\,dz = \int_0^1 (1+i)t(1+i)\,dt = \int_0^1 2it\,dt = i.$$

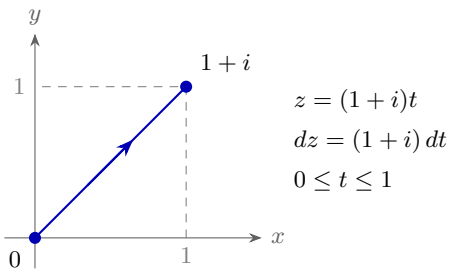


Figure 1.14: A straight contour from 0 to $1+i$. The arrow fixes the orientation; the complex tangent is the constant $1 + i$.

1.4.4 Orientation, addition, and a circular contour

Contour integration is linear in the integrand. It is unchanged by a piecewise smooth, orientation-preserving reparametrization of the same traversal. If $-\gamma$ denotes the reverse traversal and $\gamma_1 * \gamma_2$ means first following γ_1 and then γ_2 (with matching endpoints), then

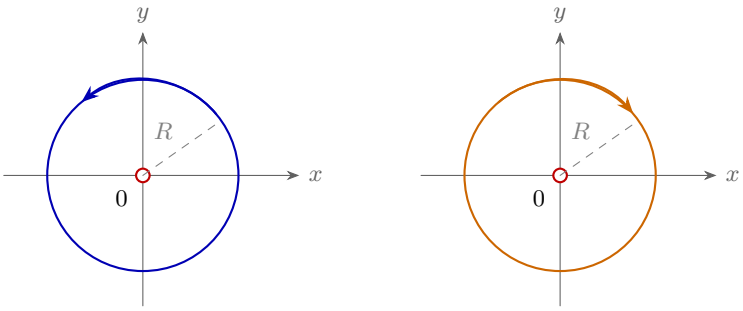
$$\int_{-\gamma} f(z)\,dz = -\int_\gamma f(z)\,dz, \qquad \int_{\gamma_1*\gamma_2} f(z)\,dz = \int_{\gamma_1} f(z)\,dz + \int_{\gamma_2} f(z)\,dz.$$

Example 12. *Evaluate $\displaystyle\oint_{C_R} \frac{1}{z}\,dz$, where C_R is the circle $|z| = R$, $R > 0$, traversed once counterclockwise.*

Solution 7. *Parametrize by $z = Re^{it}$, $0 \le t \le 2\pi$, so that $dz = iRe^{it}\,dt$. Therefore*

$$\oint_{C_R} \frac{1}{z}\,dz = \int_0^{2\pi} \frac{iRe^{it}}{Re^{it}}\,dt = \int_0^{2\pi} i\,dt = 2\pi i.$$

Clockwise traversal gives $-2\pi i$. Although $1/z$ is undefined at the center, it is continuous at every point of the contour, so the integral is well-defined. Its value does not depend on R.



Counterclockwise: $2\pi i$ Clockwise: $-2\pi i$

Figure 1.15: Reversing a contour reverses the sign of $\int (1/z)\,dz$. The excluded point 0 lies inside each circle, not on the path.

1.4.5 Path dependence and antiderivatives

Example 13. *Compare $\int_\gamma \bar{z}\,dz$ along two paths from 0 to $1+i$: the diagonal segment γ_d, and the broken path γ_b going from 0 to 1 and then from 1 to $1+i$.*

Solution 8. *On the diagonal, $z = (1+i)t$, $0 \le t \le 1$, and hence*

$$\int_{\gamma_d} \bar{z}\,dz = \int_0^1 (1-i)t(1+i)\,dt = \int_0^1 2t\,dt = 1.$$

On the horizontal part of the broken path, use $z = t$, $0 \le t \le 1$:

$$\int_0^1 t\,dt = \frac{1}{2}.$$

On the vertical part, use $z = 1+it$, $0 \le t \le 1$, so $dz = i\,dt$:

$$\int_0^1 (1-it)i\,dt = \int_0^1 (t+i)\,dt = \frac{1}{2} + i.$$

Thus $\int_{\gamma_b} \bar{z}\,dz = 1+i$, whereas $\int_{\gamma_d} \bar{z}\,dz = 1$. The endpoints alone do not determine a contour integral.

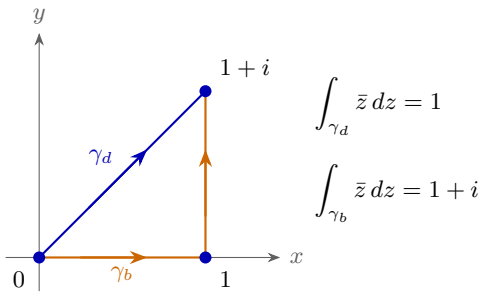


Figure 1.16: The same endpoints can give different integrals. Following γ_b and then the reverse of γ_d forms a counterclockwise triangular loop, whose integral of $\bar{z}$ is $(1+i) - 1 = i$.

Definition 12. *An* antiderivative *(or* primitive*) of f on an open set $D \subseteq \mathbb{C}$ is an analytic function $F : D \to \mathbb{C}$ such that $F'(z) = f(z)$ for every $z \in D$.*

Theorem 2 (Fundamental theorem for contour integrals)**.** *Suppose f is continuous on an open set D and has a primitive F there. For every contour $\gamma : [a, b] \to D$,*

$$\boxed{\int_\gamma f(z)\,dz = F(\gamma(b)) - F(\gamma(a)).}$$

Consequently, the integral depends only on the endpoints, and the integral around every closed contour in D is zero.

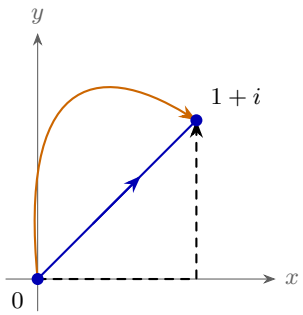
Proof. On each smooth piece the chain rule gives $(F \circ \gamma)'(t) = F'(\gamma(t))\gamma'(t) = f(\gamma(t))\gamma'(t)$. Integrating and adding the endpoint differences over the pieces proves the formula. □

Example 14. *Evaluate $\int_\gamma z\,dz$ along any contour from 0 to $1+i$, and compare with the straight-segment calculation.*

Solution 9. *The function $F(z) = z^2/2$ is a primitive of z on all of $\mathbb{C}$. Therefore, regardless of the chosen contour,*

$$\int_\gamma z\,dz = \frac{(1+i)^2}{2} - 0 = i.$$

This agrees with the earlier parametrization. In particular, the diagonal and broken paths in Figure 1.16 both give i for this integrand, even though they give different values for the integrand $\bar{z}$.



$$F(z) = z^2/2$$

$$\int_\gamma z\,dz = F(1+i) - F(0) = i$$

Figure 1.17: A primitive makes the integral independent of the path. All three paths give the same integral of z, because $z^2/2$ is defined and analytic throughout the plane.

Remark 6. *Analyticity on a domain alone does not guarantee a primitive on that entire domain. The function $1/z$ is analytic on $\mathbb{C} \setminus \{0\}$, but its integral around C_R is $2\pi i$, so it has no primitive there. On the slit plane $\mathbb{C} \setminus (-\infty, 0]$, however, $\operatorname{Log} z$ is a primitive of $1/z$. A full circle around 0 is not contained in this slit plane, so the endpoint formula using* Log *cannot be applied to that circle.*

1.4.6 Estimating integrals: the ML inequality

Theorem 3 (ML inequality)**.** *Let f be continuous on a contour γ of length L. If $|f(z)| \le M$ at every point of the contour, then*

$$\boxed{\left|\int_\gamma f(z)\,dz\right| \le ML.}$$

Proof. Using the parametrized integral and the triangle inequality gives

$$\left|\int_\gamma f(z)\,dz\right| \le \int_a^b |f(\gamma(t))|\,|\gamma'(t)|\,dt \le M\int_a^b |\gamma'(t)|\,dt = ML.$$

□

Example 15. *Find an upper bound for $\left|\displaystyle\int_\gamma \frac{1}{z-2}\,dz\right|$, where $\gamma(t) = e^{it}$, $0 \le t \le \pi$, is the upper unit semicircle from 1 to -1.*

Solution 10. *On γ, $|z| = 1$, so the reverse triangle inequality gives*

$$|z-2| \ge 2 - |z| = 1, \qquad \left|\frac{1}{z-2}\right| \le 1.$$

We may take $M = 1$. Since $|\gamma'(t)| = |ie^{it}| = 1$, the length is $L = \int_0^\pi 1\,dt = \pi$. Hence

$$\left| \int_\gamma \frac{1}{z-2}\,dz \right| \le \pi.$$

This is an estimate, not a claim that the integral has modulus π.

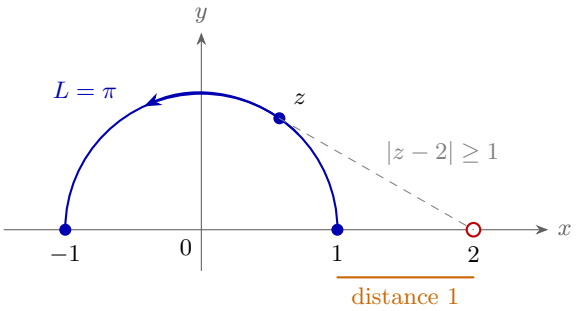


Figure 1.18: The pole at 2 stays at least distance 1 from the upper unit semicircle. This controls the size of the integrand, while the arc length supplies the other factor in the ML bound.

When working with a new contour integral, first check the path and its orientation, and check that the integrand is defined on the path. Then choose between direct parametrization, a primitive valid on a domain containing the whole path, or an ML estimate if only a bound is needed.